

SHRINKAGE ESTIMATION OF STOCHASTIC DISCOUNT FACTORS IN ASSET PRICING

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OUTLINE

1. Motivation
2. Background & Literature Review
3. Methodology
4. About the Data
5. Replication of Kozak et al. (2020)
 1. 50 Anomaly Portfolios
6. Extensions
 1. Emerging Markets vs Developed Markets
 2. Nasdaq-100 (QQQ) ETF
7. Applications
8. Conclusions & Future Work

MOTIVATION

- The goal of asset pricing is to explain why different stocks get different returns to help inform investment decisions.
- Researchers have documented 380+ firm characteristics that appear to predict stock returns, referenced as the “factor zoo” [3].
- Traditional asset pricing models (CAPM, Fama-French, etc.) use a handful of these characteristics and assume they can accurately describe investment return.
- In practice, many of these factors contain critical pricing information.
- A Stochastic Discount Factor (SDF) allows us to consider (and weight) many characteristics according to their contribution to explaining cross-sectional return differences.
- The core challenge is **estimating the SDF coefficients**.

BACKGROUND – THE STOCHASTIC DISCOUNT FACTOR (SDF)

- The SDF is expressed as: $M_t = 1 - b'(F_t - E[F_t])$,
 - b is an $H \times 1$ vector of SDF coefficients (what we want to estimate).
 - F_t is an $H \times 1$ vector of excess returns on the H characteristics.
 - M_t is the stochastic discount factor: A random variable that adjusts the price of assets by adjusting for risk across economic states.

Pricing condition: $E[M_t F_t] = 0$

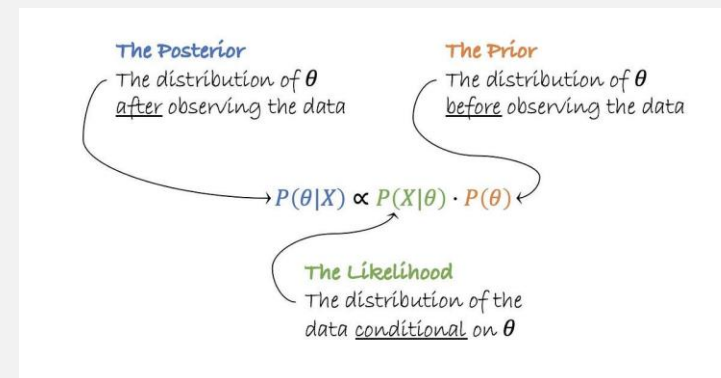
- This pricing condition associates the stochastic discount factor to an asset price. Setting the pricing condition equal to 0 means we don't have arbitrage (the existence of riskless investments with positive expected returns)
- Implied expected excess return for any asset i : $E[F_{i,t}] = Cov(F_{i,t}, b' F_t)$

LITERATURE REVIEW

- Traditional approach: hand-select a few factors (e.g., Fama-French 3-factor)
- Modern reality: “Factor Zoo” with 380+ characteristics
- Naïve OLS estimator
 - $\hat{b}_{ols} = \bar{\Sigma}^{-1} \bar{\mu}$
- Key Paper: Kozak, Nagel, & Santosh (2020) “Shrinking the cross-section”
 - Argues the cross-section is not sparse
 - Propose Bayesian shrinkage with economically motivated prior
 - Prior penalizes implausible high Sharpe ratios in low-variance directions
- Kozak et al. (2020) innovation: develop a clean, IID prior on b .

BAYESIAN SHRINKAGE: PRIOR, POSTERIOR, & ESTIMATOR

- Prior on expected returns μ is $\mu \sim N(0, \frac{\kappa^2}{\tau} \Sigma^\eta)$, with $\eta = 2$
- Likelihood (data): $\bar{\mu}|\mu \sim N(\mu, \frac{1}{T} \Sigma)$
- After decomposing $\Sigma = QDQ'$ (and focusing on D , the eigenvalues) we get an IID prior on SDF coefficients. Recall $\mu = \Sigma b$.
 - $b \sim N(0, \frac{\kappa^2}{\tau} I)$
- The normal prior on μ and normal likelihood for $\bar{\mu}$ form a conjugate pair, giving a closed form posterior for b .
- The final SDF estimator used in practice is the posterior mean:
 - $\hat{b} = (\Sigma + \gamma I)^{-1} \bar{\mu}$, where $\gamma = \frac{\tau}{\kappa^2 T}$
- The posterior variance is $\text{var}(b|y) = \frac{1}{T} (\Sigma + \gamma I)^{-1}$



ALGORITHM

1. Input data (monthly excess returns F_t for H portfolios over T months).
2. Calculate sample statistics, $\bar{\mu}$ and $\bar{\Sigma}$.
3. Choose κ based on what minimizes the out-of-sample R^2 . κ determines the optimal amount of shrinkage necessary.
 1. $R_{\text{oos}}^2 = 1 - \frac{(\bar{\mu}_2 - \bar{\Sigma}_2 \hat{b})' (\bar{\mu}_2 - \bar{\Sigma}_2 \hat{b})}{\bar{\mu}_2' \bar{\mu}_2}$.
4. Apply Bayesian Shrinkage Estimator:
 1. Compute $\gamma = \text{frequency} \times \frac{\tau}{\kappa^2 T}$.
 2. Compute the shrunk SDF coefficients: $\hat{b} = (\Sigma + \gamma I)^{-1} \bar{\mu}$.
5. Output: $\hat{b}$, which is the stochastic discount factor.

REPLICATION

ABOUT THE DATA

- This study relies on publicly available datasets drawn from the Kenneth French Data Library [5] and data from Financial Modeling Prep.
- Key Properties
 - Monthly returns
 - Value weighted
 - Returns calculated in excess of one month Treasury Bill rate (used to describe standard market returns)
 - Returns calculated with the US dollar
- Returns based on long-short portfolios sorted on different characteristics.

Wall Street Prep

Long/Short Equity Fund

Long Position ↑

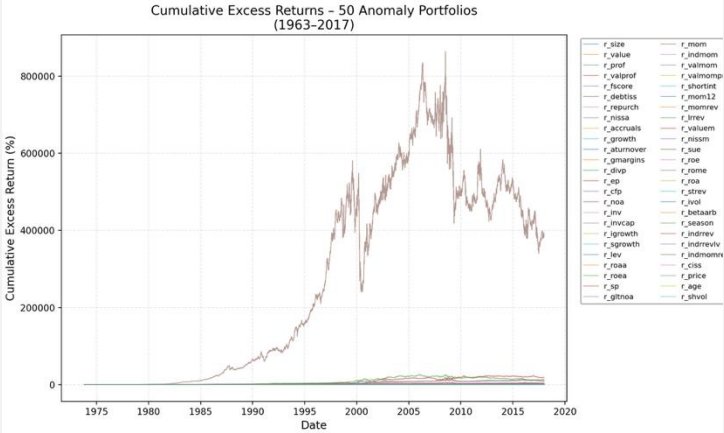
Equities anticipated to rise in share price are purchased to profit from the upside

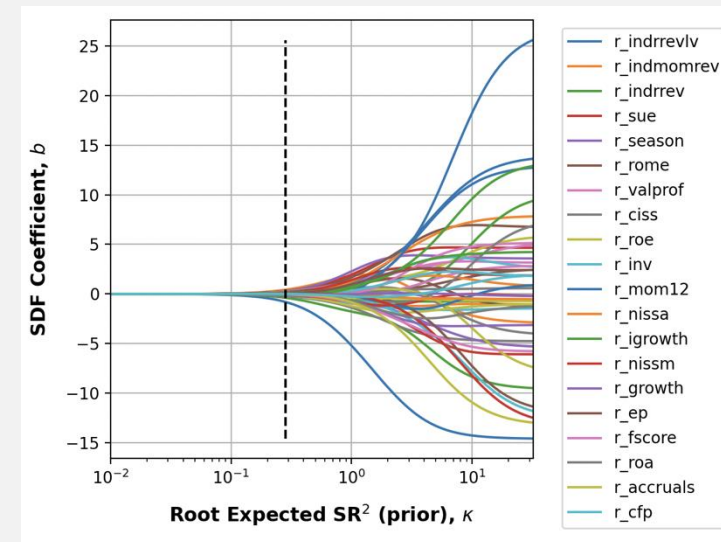
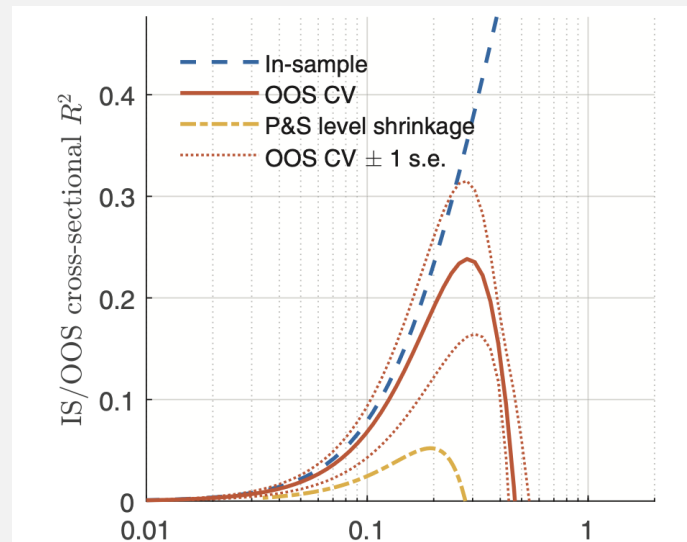
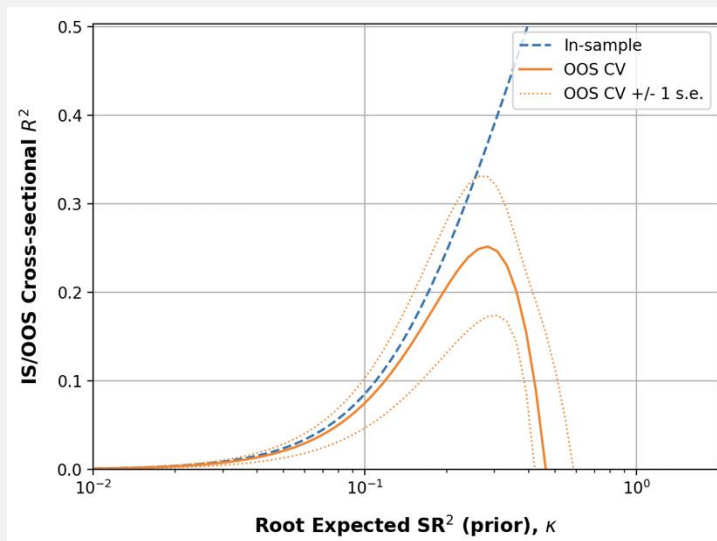
Short Position ↓

Securities borrowed from a brokerage are sold to profit from repurchasing the securities at a lower price

50 ANOMALY PORTFOLIOS

- 50 long-short strategies based on well-known firm characteristics (momentum, profitability, accruals, volatility, etc.)
- Specifically designed to challenge traditional models (like CAPM)
- Returns are orthogonalized to the market factor to focus on cross-sectional variation.
- Moderately high dimensional setting ($H = 50$) with significant correlation between anomalies.
- Serves as a realistic and difficult test case for shrinkage methods.





50 ANOMALY PORTFOLIOS

- 3-fold cross-validation selects the optimal shrinkage parameter, $\kappa \approx 0.3$, which matches the results for this dataset in Kozak et al. (2020).
- The shrunk SDF concentrates on a smaller number of signals: industry relative reversal, momentum reversal, unexpected earnings (SUE), etc.
- Most other anomalies are heavily shrunk towards zero.
- Out-of-sample R^2 improves significantly compared to naïve OLS estimator.

EXTENSIONS

DEVELOPED MARKETS VS EMERGING MARKETS DATA

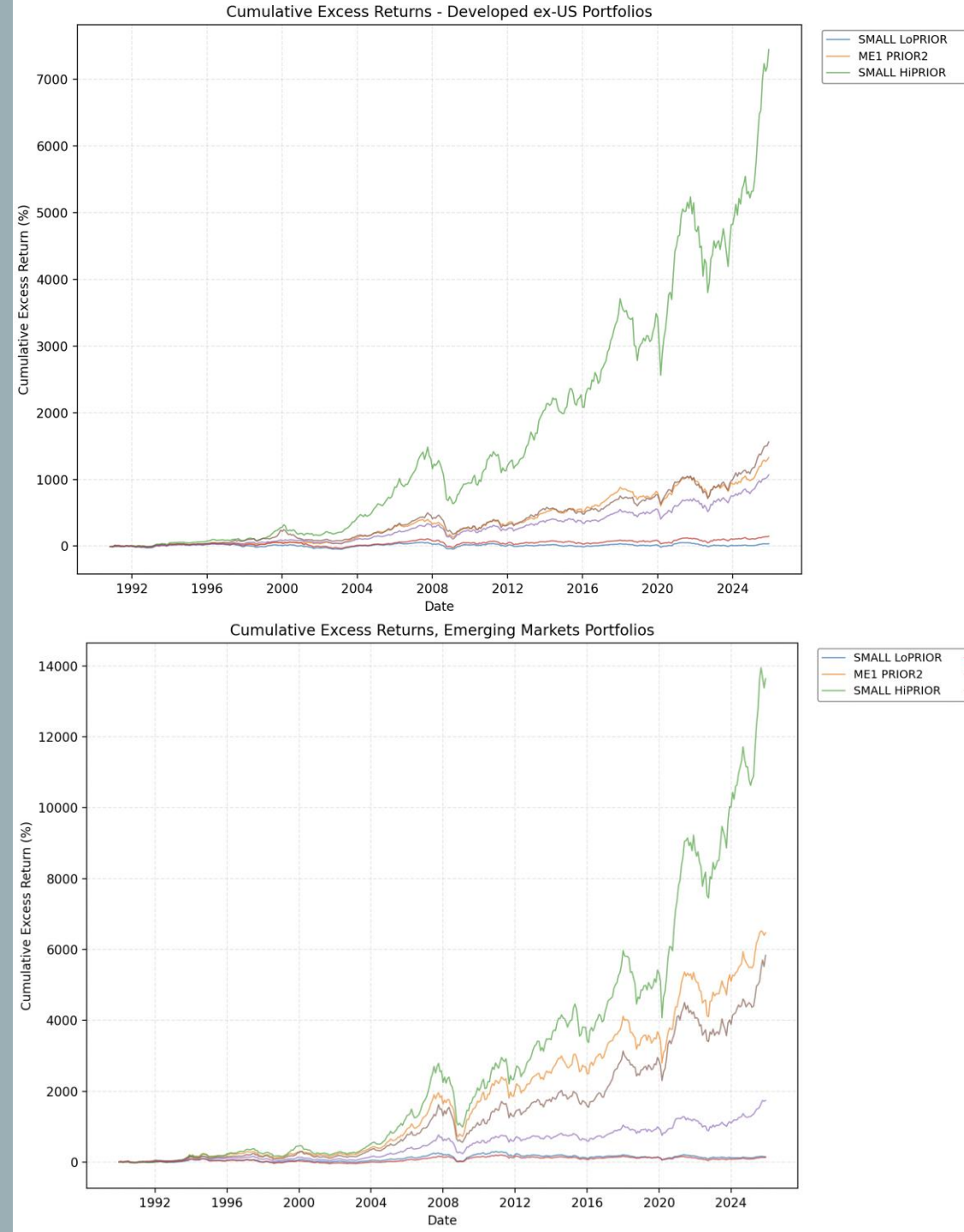
Two international size-momentum datasets from Kenneth French Data Library.

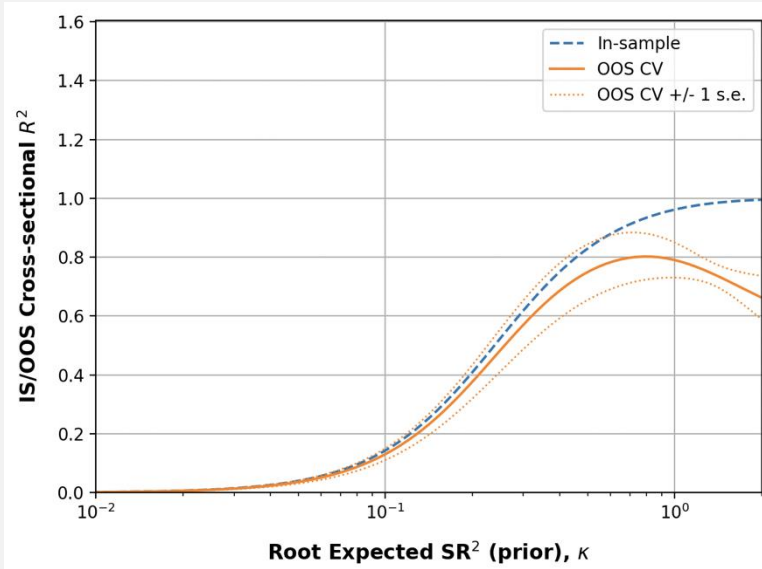
Developed markets (ex-US): More stable, lower volatility, mature economies.

Emerging markets: Higher volatility, greater return dispersion, noisier signals.

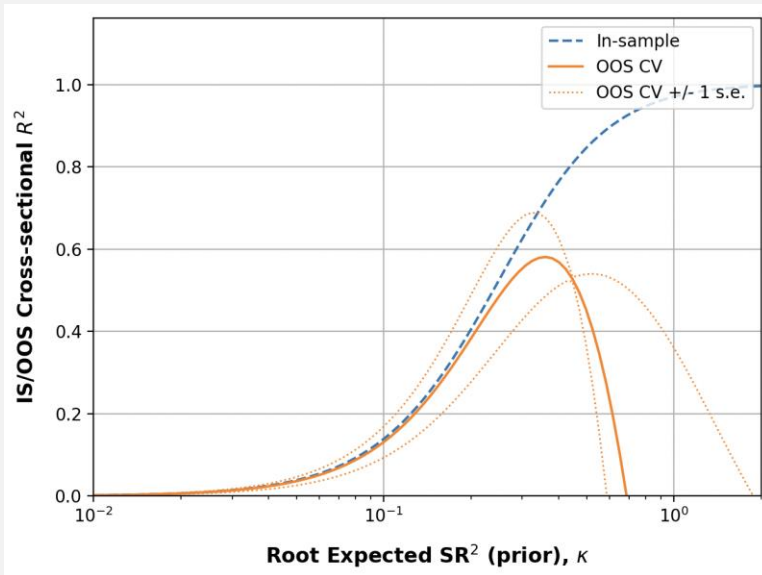
Both datasets use size and momentum sorts for the long-short portfolios.

Goal: test how our shrinkage estimator performs across different economic regimes.





(a) Developed Markets ex-US



(b) Emerging Markets

κ OPTIMIZATION

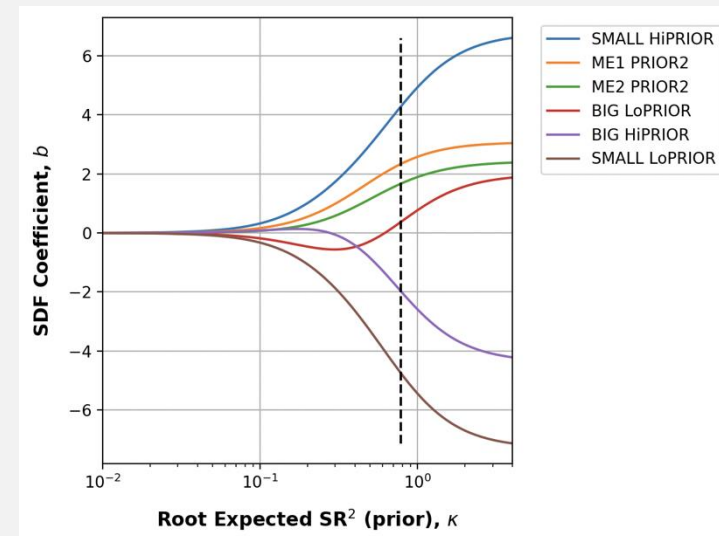
We use 3-fold cross-validation to select the optimal shrinkage parameter, κ , for both the emerging markets data and the developed markets ex-US data.

Plot (a): Flatter R_{OOS}^2 , more stable return patterns, optimal κ indicates we benefit from moderate shrinkage.

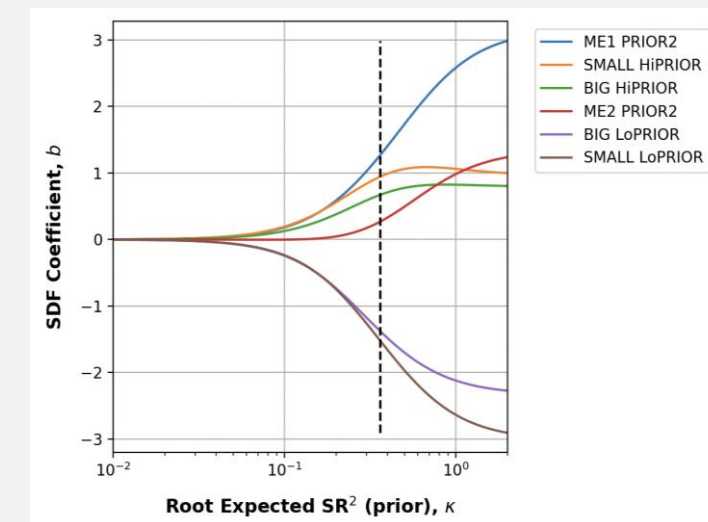
Plot (b): Sharper decline in R_{OOS}^2 after the peak, higher volatility and greater return dispersion, requires stronger shrinkage (lower optimal κ)

SDF RESULTS

- Both plots show how the SDF coefficients, b , change as shrinkage strength, κ , varies.
- Plot (a): Developed Markets (ex-US)
 - Larger range on the y-axis demonstrates a larger coefficient magnitude compared to emerging markets
 - Strong loadings on momentum signals
 - Small LoPRIOR and Small HiPRIOR dominate (which are opposite signs)
 - More stable signals
- Plot (b): Emerging Markets
 - Smaller range on the y-axes demonstrates smaller coefficient magnitudes
 - Momentum related portfolios still dominate, but with more noise
 - Loadings are weaker overall due to higher volatility and dispersion



(a) Developed Markets ex-US



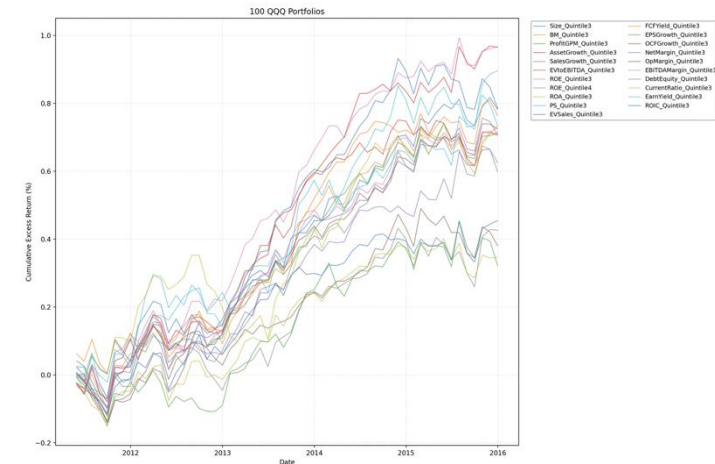
(b) Emerging Markets

EMERGING MARKETS VS DEVELOPED MARKETS (EX-US) KEY TAKEAWAYS

- Both markets show clear benefits from Bayesian shrinkage (positive optimal κ)
- Developed Markets (ex-US)
- Emerging Markets
 - Smaller coefficient magnitudes
 - Momentum dominates, but with more noise
 - Requires stronger shrinkage due to volatility
- Main Insight
 - The shrinkage estimator adapts flexibly to different economic regimes
 - Stronger, more reliable signals in developed markets
 - Still extracts meaningful pricing information in noisier emerging markets

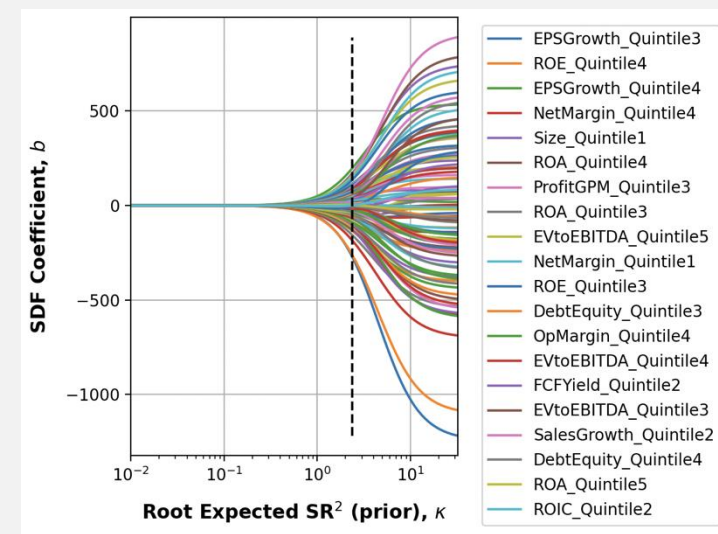
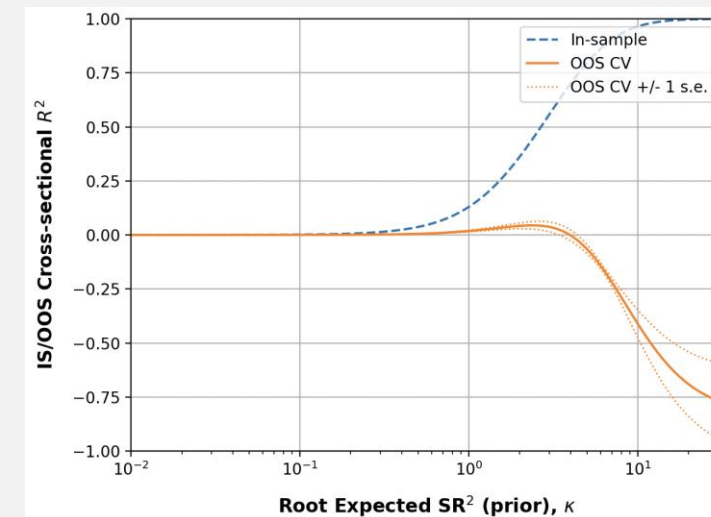
QQQ ETF DATA

- Custom high-dimensional dataset constructed from Financial Modeling Prep using monthly data from the Nasdaq-100 QQQ ETF.
- 20 fundamental, valuation, profitability, and growth characteristics are used to create long-short portfolios: size, book-to-market, ROE, EPS growth, net margin, EV/EBITDA, etc.).
- QQQ focuses on a narrow large-cap growth/tech universe.
- Stocks sorted into 100 value weighted portfolios (single-sorted).
- Using monthly data over 10 years (120 observations) creates a short time series when using 3-fold cross-validation, making this a challenging test case for shrinkage.
- Goal: stress-test the shrinkage estimator in a highly correlated, shorter-sample environment



κ OPTIMIZATION & SDF COEFFICIENTS

- κ optimization shows a modest, but positive, peak in the R_{OOS}^2 .
- Optimal shrinkage is relatively small (due to limited historical data and the nature of ETFs).
- SDF loadings are dominated by profitability and growth signals.
 - Strong negative loadings on moderate EPS growth and high ROE.
 - Positive loadings on higher EPS growth and smallest stocks within QQQ.
- Many characteristics heavily shrunk due to high collinearity/redundancy.
- Key takeaway: the SDF coefficients reveal strong loadings on EPS growth and ROE in the QQQ universe, but overall out-of-sample performance is limited by high correlations and shorter sample length.



APPLICATIONS

- The estimated SDF coefficients, $\hat{b}$, can be interpreted to make practical stock selections.
- For any stock i , we can compute a score based on its characteristics:

$$Score_i = \sum_j \hat{b}_j \times z_{ij}$$

- z_{ij} is the quintile value of characteristic j for stock i . $\hat{b}_j$ is the SDF coefficient for characteristic j .
- A higher score means a stock is expected to deliver higher excess returns across the model. Invest in these stocks!
- Lower scores imply a stock is expected to underperform. Underweight or avoid these stocks.
- Example from QQQ Data:
 - Strongly underweight stocks with moderate EPS growth (Q3) and high ROE (Q4).
 - Overweight stocks with high EPS growth, very low net margin, and the smallest names within the large-cap universe.
- Practical Value: Converts abstract SDF weights into a simple, transparent stock scoring system.

CONCLUSION & FUTURE WORK

- Main contribution:
 - Successfully replicated and extended Kozak et al. (2020) Bayesian shrinkage estimator for SDF coefficients.
- Key Findings:
 - Moderate shrinkage consistently improves out-of-sample cross-sectional R^2 .
 - The method effectively makes large sets of correlated characteristics sparse.
 - Performs well across US, developed, and emerging markets, with natural adaptation to different noise levels.
 - In the challenging QQQ growth universe, it identifies profitability and earnings growth as the dominant signals
- Future work:
 - Expand to even larger characteristic sets or daily-frequency data.
 - Develop end-to-end portfolio optimization framework using the shrunk SDF.

SUPPLEMENTARY SLIDES

THE USE OF κ IN THE BAYESIAN ESTIMATOR

- Choose κ by what optimizes the out-of-sample R^2 :
 - $R_{OOS}^2 = 1 - \frac{\bar{\mu}}{\tau}$
- $\gamma = freq \times \frac{\tau \kappa^2}{T}$

ECONOMIC MOTIVATION FOR PRIORS

- Why choose $\eta = 2$
- Multivariate normal conjugate pair
- switching from μ to D

TRADITIONAL CAPM MODEL

- Maybe a simple example of what a CAPM model looks like

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